Design Review request form

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1. SCOPE

Aim of this document is to collect basic information on the customer application that will help the Telit Design Review team in having a clear understanding of the application constraints and requirements of the whole application environment. This information together with the information provided for the Design Review (Schematic, PCB Gerber, PCB layup) will assist Telit Design Review team in providing customer with a tailored Design Review Report.

2. INSTRUCTIONS

Please fill this document in all the parts either checking the options or adding text description. For your convenience we repeat here the list of files required for a complete Design Review.

Please check the correctness of the documents before sending.

We need the following information:

1. Schematic files:

    1.1. PDF with searchable names of components and signals.

    1.2. Part name and description or BOM file.

2. CAD files:

    2.1. Gerber files in format RS247X

    2.2. Drill files in excellon format with preferred separate files for plated and non plated vias.

    2.3. Netlist file in format IPC-356A.

    2.4. PCB layout description with copper and dielectric width, dielectric constant.

    2.5. PCB stack-up, description and gerber filename correspondences organized from top to bottom: layers, silkscreens, pastes and solder masks.

    2.6. In alternative to gerbers files, the full single ODB++ zip file.

In order to speed up the Design Review for subsequent DR revisions:

The subsequent design review revisions requests must come with a change log description and Review ID.

3. CUSTOMER İNFORMATİON

Company name :	Ovoo Electronics
Country :	Turkey
Address :	Şeker Mah. Ayyıldız Sok. Silverstar Apt. No:7/7 42050 Karatay / Konya
Contact Name :	Mehmet Günce Akkoyun
Contact e-mail :	akkoyun@me.com

4. GENERAL İNFORMATİON

Application name :	B101CA Rev 00.00.03 : qSlot / PCIe Form Factor GSM Modem Module
Telit module integrated in the application :	LE910C1-EUX
Field of application :	Smart agriculture and industrial power monitoring systems.
Installation description :	Module is installed in electricity boxes for measuring electric parameters.
Other radio device integrated in the application :	-
Other useful information :	The application uses an external cellular antenna connected directly to the onboard RF connector. The board is powered from a single-cell Li-Ion battery / system power rail. The module is intended for outdoor electrical panel installations used in smart agriculture and industrial power telemetry systems. This submission is a second design review request after Telit DR-35567 Rev 1. The schematic and PCB layout have been revised according to the findings in the first review.
Design Review already done on this application	Yes
Previous Design Review ID	DR-35567 Rev 1
Non Disclosure Agreement Signed :	Yes

5. GENERAL VIEW

Scope of operation – main purpose of product

-	Asset Tracking
-	Automated Meter Reading / Management
-	Automotive - Car Makers & Tier1 Supplier
-	Fleet Management - Tolling - PAYD
-	GSM Gateway
-	GSM Railway
X	Industrial Automation
-	Information & Displays
-	Mobile Computing
-	Navigation
-	POS - Point Of Sales Terminals
-	Public Transportation
-	Security - Alarm - Surveillance
-	Sub-System & Terminals based on Telit modules
-	Telemedicine / Medical
X	Telemetry
-	Telephony
-	V24 to GPS converter
-	Vending
-	Wearable Electronics
X	Other : Electrical parameter measurement and telemetry.

6. LOCATION, AMBIENT, CONDITIONS

Location :	
-	Indoor
X	Outdoor
-	Automotive
X	Other : In electric box at outdoor.

Ambient :	
X	Countryside
-	Urban

Conditions :	
-	Explosive Atmosphere
X	Extreme Conditions (specify) : Outdoor electrical control panels, typically installed in agricultural irrigation and industrial power monitoring applications. The product may operate near transformers, contactors, motor starters, VFDs, compensation panels and high-current power cables. High electromagnetic and magnetic field exposure is possible.

Installations :	
X	Fixed
-	Mobile
-	Portable

7. BANDS OF OPERATION

Required Bands :	
-	CDMA 450
-	CDMA 800
-	CDMA 1900
-	GSM 850/1900
X	GSM 900/1800
-	UMTS B1
-	UMTS B2
-	UMTS B3
-	UMTS B4
-	UMTS B5
-	UMTS B8
X	LTE I (2100)
X	LTE II (1900)
X	LTE III (1800)
X	LTE IV (1700-2100)
X	LTE V (850)
X	LTE VII (2600)
X	LTE XIII (700)
X	LTE XX (800)
X	Others : All operators in Turkey and all 2G and 4G compatible countries.

8. APPROVALS REQUIRED

Regulatory Approvals :	
X	CE Mark
X	GCF
X	FCC/IC
X	PTCRB
-	Other : _____

Carrier Approvals :	
-	AT&T
-	Verizon
-	Vodafone
X	Other : All operators in Turkey

Other Approvals :	
-	ATEX/IECEx (specify) : _____
-	Other : _____

9. ADDITIONAL RADIOS

Additional Radio Technology :	
-	WiFi
-	GPS
-	Bluetooth / BLE
-	Short Range (specify) _____

Coexistence Management :	
-	None
-	No Simultaneous Operation
-	Other : _____

10.GNSS

-	GPS
-	GLONASS
-	Galileo
-	QZSS
-	Beidou
-	Other : _____

11.ANTENNAS

-	Embedded
-	Internal
X	External, Placed directly on connector or close to application
-	External, Connected with long coax cable and placed away
-	GNSS antenna passive
-	GNSS antenna active

12.POWER

Type of power supply / battery (specify)	Single-cell Li-Ion / LiPo battery powered system. Nominal battery voltage: 3.7 V Battery operating range: approximately 3.0 V to 4.2 V Planned capacity: 2000 mAh or higher The modem power rail is generated on board. Power sequencing and PWRMON/VAUX controlled level translation were revised after DR-35567 Rev 1.
Modem RTC hold time requirement (specify)	Internal (there is no RTC backup need for modem)

13.SIM

X	Standard SIM Holder
-	Standard SIM Holder / SIM on Chip Mount Alternative
-	SIM on Chip Only
-	Dual SIM
-	Telit IoT Connectivity
-	SimWise (SIM profile embedded in Module)

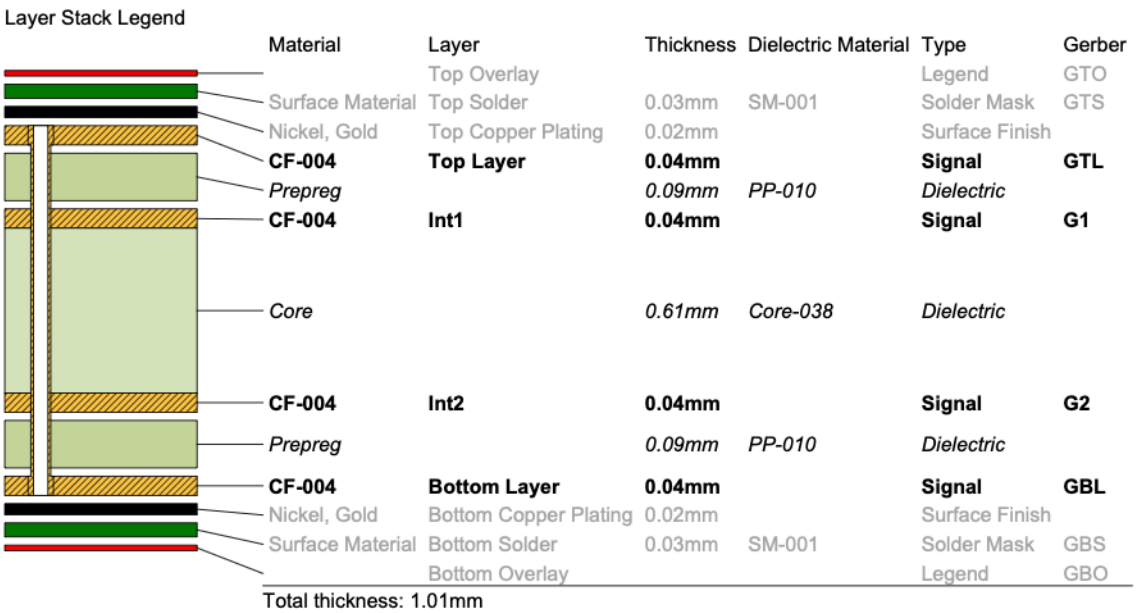
14.MODULE CONTROL

-	Embedded Telit Python Script
-	Embedded Telit IoT AppZone
X	External Microcontroller or Interface

15.DOCUMENTATION FILES TABLE

Schematic Files			
File Type	File Name	Format	Note
Schematic	B101CA_00.00.03_Schematic.PDF	PDF	
BOM	B101CA_00.00.03_BOM.CSV	PDF or Excel	
Custom Components Datasheet		PDF	Antennas, custom components etc.

PCB Stack-up [4 Layer]



PCB Extra Files				
File Type	File Name	Format	From Layer	To Layer
Silkscreen Top				
Paste Mask Top				
Solder Mask Top				
Copper 1 [Top]				
Dielectric 1				
Copper 2 [Inner 1]				
Dielectric 2				
Copper 3 [Inner 2]				
Dielectric 3				
Copper 4 [Bottom]				
Solder Mask Bottom				
Paste Mask Bottom				
Silkscreen Bottom				

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Mod.0363 12-2016 Rev.6

DESC.

- **This is a second design review request for the revised B101-BA modem board.**

The previous Telit design review report was DR-35567 Rev 1. The design has been updated according to the findings in that report.

Summary of implemented changes:

1. The LE910C1-EUX modem footprint was corrected according to the Telit recommended footprint.

2. Reserved pins K4 and M6 were disconnected and marked as NC.

3. The voltage translator VCCB supply was changed from VIO_1V8 to PWRMON/VAUX to reduce latch-up risk.

4. L3 inductor information was clarified in the schematic as a shielded / magnetic-resin-shielded power inductor.

5. The RF 50 Ohm waveguide structure was preserved.

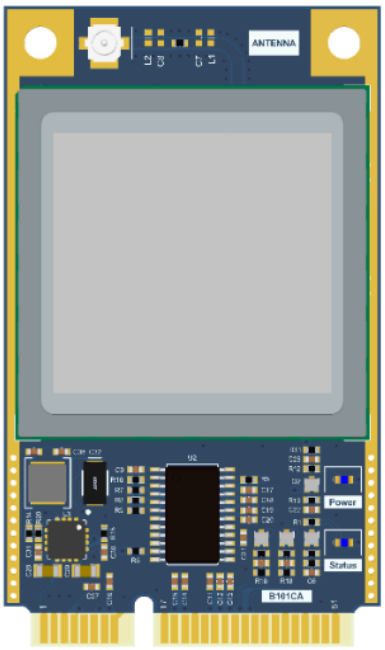
6. SIM interface design was preserved because it was marked OK in the previous review.

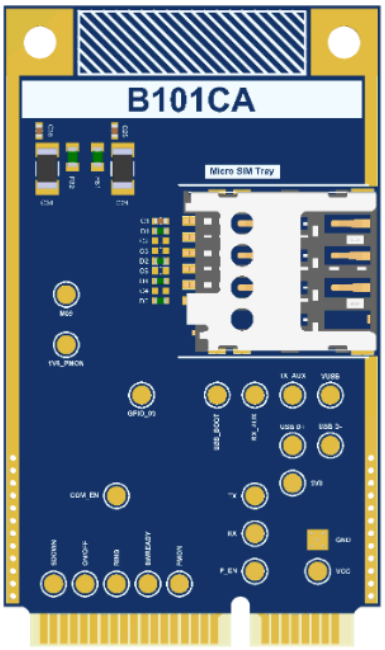
Open / under evaluation:

1. Fast shutdown implementation is under evaluation. Required hold-up time, power-fail detection and firmware shutdown sequence are being reviewed.

2. Optional 33 pF COG/NPO DNP filter capacitor footprints for RF/harmonic suppression are under evaluation for EMC preparation.

The PCIe card-edge form factor is used only for mechanical integration. The electrical pinout is custom and belongs to our qSlot interface. It is not intended to be PCIe compatible. Detail : https://www.akkoyun.net/acik-kaynak/qslot





16.CHANGE LOG

Change Log	
Revision	Description
00.00.01	First design submitted for Telit design review.
00.00.03	Second design review submission after Telit DR-35567 Rev 1. Implemented changes: - Corrected LE910C1-EUX modem footprint according to Telit recommended footprint. - Disconnected reserved pins K4 and M6; both pins are now NC. - Changed voltage translator VCCB supply from VIO_1V8 to PWRMON/VAUX. - Clarified L3 as shielded / magnetic-resin-shielded 2.2 µH inductor in schematic and BOM. - Preserved the 50 Ohm RF waveguide, previously marked OK. - Preserved SIM interface, previously marked OK. - Updated schematic, PCB layout, BOM and Gerber files for Rev 00.00.03. Open items: - Fast shutdown feature is under evaluation. - Optional 33 pF DNP RF/harmonic filter capacitor footprints are under evaluation.